

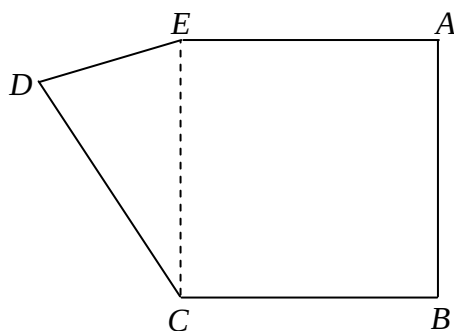
- 1 Sketch the graphs of $y = 3|x+1|$ and $y = \frac{1}{4}x^3 - \frac{1}{2}x^2 - 6x + 15$ on the same diagram, and hence, solve the inequality $|x+1| \leq \frac{1}{12}x^3 - \frac{1}{6}x^2 - 2x + 5$. [4]

- 2 Given that $y^2 + e^x + 3y = 5$, $y > 0$,

(i) show that $2\left(\frac{dy}{dx}\right)^2 + (2y+3)\frac{d^2y}{dx^2} + e^x = 0$, [2]

(ii) find the first three terms of the Maclaurin's series for y . [3]

- 3 The figure below shows the floor area of a room. $ABCE$ is a square of side 2 metres. ECD is a triangle with $CD = CE$ and angle $DCE = \theta$ radians. The floor costs \$8000 to carpet. Given that θ is a sufficiently small angle, show that the unit cost of carpeting is $\$(a + b\theta)$ per square metre, where a and b are constants to be determined. [5]



- 4 Find

(i) $\int \sec^3 x \tan x \, dx$, [2]

(ii) $\int \frac{1}{2\sqrt{x}(x-9)} \, dx$, by using the substitution $u = \sqrt{x}$. [4]

- 5 The planes Π_1 and Π_2 have equations

$$\mathbf{r} \cdot (\mathbf{i} - 3\mathbf{j} + 2\mathbf{k}) = 6 \quad \text{and} \quad 2x - y - z = 2,$$

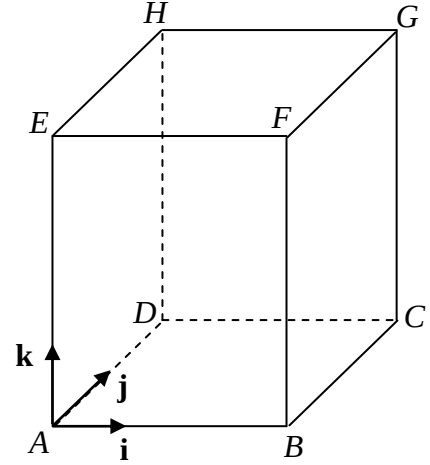
respectively. The point P with position vector $2\mathbf{i} + 2\mathbf{k}$ lies on Π_1 .

- (i) Verify that P lies on Π_2 . Hence, or otherwise, find a vector equation of the line of intersection between Π_1 and Π_2 . [3]

The plane Π_3 has equation $x + by - z = 4b$, where b is a real constant.

- (ii) When $b = -3$, find the position vector of the point of intersection of the 3 planes. [1]
 (iii) Determine the value(s) of b such that the 3 planes intersect at a common line. [2]

- 6 The diagram shows a rectangular cuboid with $AB = 4$ units, $AD = 3$ units and $AE = 5$ units. The point A is taken as origin and the unit vectors $\mathbf{i}$, $\mathbf{j}$ and $\mathbf{k}$ are parallel to AB , AD and AE respectively. M is the mid-point of AH and N is a point on the line segment GH such that $HN = 3NG$.



l_1 is the line passing through the points M and N and l_2 is the line passing through the points B and H .

l_1 and l_2 intersect at the point P such that $\vec{AP} = \frac{1}{10}(12\mathbf{i} + 21\mathbf{j} + 35\mathbf{k})$.

- (i) Find the acute angle between the lines l_1 and l_2 . [3]

- (ii) Write down a vector normal to the plane $EFGH$.

Hence find the acute angle between the plane $EFGH$ and a line parallel to $\vec{AP}$. [3]

- 7 A researcher breeds fruitflies in a small jar for experimental purposes. Observations of the number of fruitflies at the end of each week suggest that the population satisfies the following recurrence relation,

$$u_{n+1} = (2 - q)u_n - 4,$$

where u_n denotes the number of fruitflies at the end of the n^{th} week and q denotes the proportion of fruitflies that die of natural causes over the week.

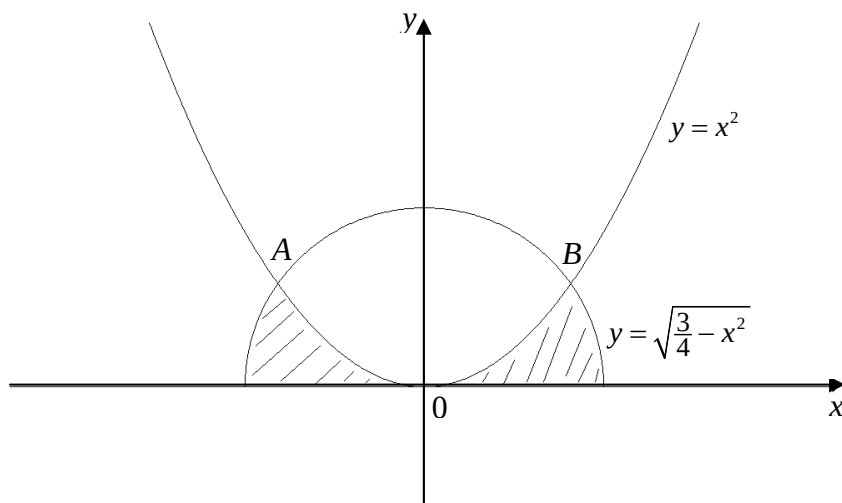
- (i) Given that $q < 1$, by considering $u_{n+1} - u_n$, show that there must initially be more than $4(1-q)^{-1}$ fruitflies in the jar for the population of fruitflies to grow. [2]
- (ii) Suppose that there are 10 and 15 fruitflies in the jar at the end of the first and second weeks respectively. Find the value of q and calculate the number of fruitflies in the jar at the end of the 20th week, giving your answer to 1 significant figure. Comment on the practicality of your answer. [4]
- (iii) Another researcher suggests using a differential equation to model the growth of the population of fruitflies instead of a recurrence relation. Should this suggestion be taken up? Explain your answer. [1]

- 8 Use the method of mathematical induction to prove that

$$\sum_{r=2}^n r^2 [(r-1)!] = (n+1)! - 2, \text{ for } n \in \mathbb{Z}, n \geq 2.$$

Hence find $\sum_{r=2}^{n-1} (r+1)^2 (r!)$. [7]

9



The diagram above shows the graphs of $y = \sqrt{\frac{3}{4} - x^2}$ and $y = x^2$. Find, in exact form, the coordinates of the points of intersection A and B. [2]

The shaded region R is bounded by the two curves and the x-axis.

- (i) Find the area of R, giving your answers correct to two decimal places. [3]
 (ii) Find the exact volume of the solid of revolution formed when R is rotated through π radians about the y-axis. [3]

- 10 The curve C has equation $y = \frac{2x^2 + ax + 3a}{x + b}$, where a and b are non-zero constants.

It is given that $x = 2$ is an asymptote of C.

- (i) State the value of b. [1]
 (ii) By differentiation, find the range of values of a for which C has no stationary points. [3]
 (iii) Given that $a = -3$, sketch C giving the equations of the asymptotes and the coordinates of any points of intersection with the axes. [3]
 (iv) State precisely a sequence of transformations which transform C to the graph of

$$y = \frac{2x^2 + ax + 3a}{3(x + b)} + 5. \quad [2]$$

- 11** In a mathematical model, the rate of change of y with respect to x is inversely proportional to $x(x+1)$, where x and y are variables with $x < 0$ and $x \neq -1$.

- (i)** Given that the rate of change of y with respect to x is $-\frac{1}{2}$ when $x = -2$, show that

$$\frac{dy}{dx} = -\frac{1}{x(1+x)} \dots \text{(I)}. \quad [2]$$

Hence, find the general solution of y in terms of x . [2]

- (ii)** Find the particular solution of (I) for which $y = \ln 3$ when $x = -\frac{1}{4}$. [1]

- (iii)** Sketch, for $x < 0$ and $x \neq -1$, the graph of the solution found in **(ii)**.
On the same diagram, sketch one other member of the family of solution curves with exactly two negative x -intercepts.

[The curves should be clearly labeled with their equations and the equation(s) of any asymptote(s).] [3]

- 12 (a)** The arithmetic progression 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, ... is arranged in rows in the following way:

1
2, 3
4, 5, 6
7, 8, 9, 10
...

The first row consists of one integer, the second row consists of the next two integers, the third row consists of the next three integers and so on.

In general, the n^{th} row consists of n integers.

- (i)** Show that the first integer in the n^{th} row is given by $1 + \frac{1}{2}n(n-1)$. [2]

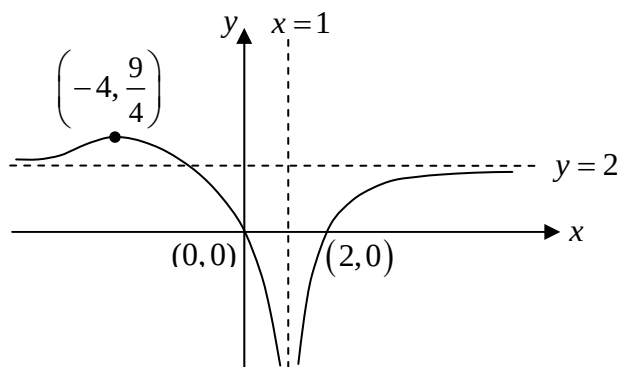
- (ii)** Determine the sum of all integers in the n^{th} row, leaving your answers in terms of n . [2]

Find the least value of n such that the sum of all the integers in the n^{th} row exceeds 10^5 . [1]

- (b)** A geometric progression has positive terms. The first term is a and the common ratio is r , where $0 < r < 1$. The sum of the first four terms of the progression is $\frac{16}{17}$ of the sum of the first eight terms.

Find the sum to infinity of the series, leaving your answer in terms of a . [4]

- 13 The diagram shows the graph of $y = f(x)$. The curve has a maximum point at $\left(-4, \frac{9}{4}\right)$ and it cuts the x -axis at the origin and at the point $(2, 0)$. The lines $x = 1$ and $y = 2$ are asymptotes to the curve.



Sketch on separate diagrams, the graphs of

- (i) $y = f(2x - 3)$, [3]
 (ii) $y = f'(x)$, [3]
 (iii) $y^2 = f(x)$. [3]

Your sketches should show clearly the equations of all asymptotes, the coordinates of all stationary points and the intersection with the axes, where applicable.

- 14 (i) The complex number z satisfies the following equation:

$$zz^* + ia(z - z^*) + 2a^2 = 2 + a(z + z^*), \quad a \in \mathbb{R},$$

where z^* is the complex conjugate of z .

- (a) By writing in the form $z = x + iy$, $x, y \in \mathbb{R}$, show that the Cartesian equation of the locus of z is $(x - a)^2 + (y - a)^2 = 2$. [2]
 (b) For $a = 1$, express the equation of the locus of z in the form $|z - v| = b$, where v is a fixed complex number and b is a real number.

Given also that $\arg(z) = \frac{\pi}{4}$, find z^* in the exponential form. Hence find the set of values of the positive integer n for which $(z^*)^n$ is purely imaginary. [5]

- (ii) Solve $(w - 1 - i)^5 = 4\sqrt{2}$, giving your answers in the form $w = q + re^{i\theta}$, where q is a fixed complex number, r is a real number, and $-\pi < \theta \leq \pi$. Give the Cartesian equation of the geometrical shape that the points represented by the roots of the equation lie on. [4]

END